MEASURE INDUSTRIAL EFFLUENT WATER pH AND ASSESS WATER QUALITY AGAINST ALLOWED STANDARDS

Exp. No. :

Date :

AIM/ OBJECTIVE:

To determine the pH of given water by pH meter and assess water quality against standards.

PRINCIPLE:

The pH of a solution is related to the H+ ion concentration by the following formula,

$$pH = -\log [H^+]$$

Measurement of pH for a solution gives the concentration of H^+ ions in the solution. When NaOH is added slowly from the burette to the solution of HCl, the fast moving H^+ ions are progressively replaced by slow moving Na^+ ions. As a result pH of the solution increases.

$$HCl + NaOH \text{ -------> } NaCl + H_2O$$

The increase in pH takes place until all the H^+ ions are completely neutralized (upto the end point). After the end point, further addition of NaOH increases pH sharply due to the fast moving OH^- ions.

PROCEDURE:

The burette is filled with the given NaOH solution. Exactly 10 ml of the given water is pipetted out into a clean beaker. It is then diluted by adding 20 ml distilled water. The glass electrode is dipped into the solution which is connected to a pH meter. Now NaOH solution is gradually added from the burette in 0.5 ml portions to the water taken in the beaker and then the pH of the solution is noted for each addition. This process is continued until at least 5 readings after the end point. The observed pH values are plotted against average volume of NaOH added. From the graph the end point is noted as V1.

OBSERVATIONS/ INFERENCE:

TITRATION OF EFFLUENT WATER Vs NaOH

Volume of water taken = ml

S.No	Volume of NaOH (ml)	pH	ΔpH	ΔV (ml)	ΔpH/ΔV	Avg. vol. of NaOH (ml)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
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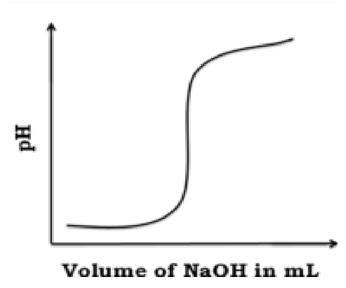
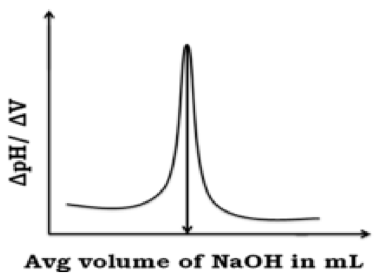
28						
29						
30						
31						

Model graph:





TITRATION OF EFFLUENT WATER Vs. NaOH

Calculation:

Volume of NaOH (V_1) = ml (from the graph)

Normality of NaOH (N_1) = N

Volume of sample water (V_2) = ml

Normality of sample water (N_2) = ?

According to the law of volumetric analysis,

$$V_1N_1 = V_2N_2$$

$$N_2 = V_1N_1/V_2$$

Normality of the sample water **(**N_2) = N

Amount of HCl present in 1 litre of the given solution =. Normality of HCl x Eq. Wt of HCl

= x 36.5

= g/L

RESULTS & DISCUSSION:

1. pH of sample water =
2. The pH of given water sample obeys /not obey allowed standard
3. Amount of HCl present in 1 litre of the solution = g/L

MAPPING OF PO & PSO: (For all the COs covered by this experiment)

CO No	PO1	PO2
1	2	1

ASSESSMENT:

Particulars	Max. Marks	Marks awarded
Preparation	10	
Conduct of Experiment	30	
Results & Discussion	30	
Viva - Voce	20	
Report	10	
Total	**100**	
Evaluator's Signature		